

Equilibrium Distribution for t-Distributed Stochastic Neighbor Embedding with Generalized Kernel

Yi Gu

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1 Introduction

The t-distributed stochastic neighbor embedding(t-SNE) is a well-known method of visualizing high-dimensional data with low-dimensional vectors (so they can be easily visualized). Let $d, s \geq 2$ with $d > s$, and $n \geq 1$. The input of t-SNE consist of vectors $X_1, \dots, X_n \in \mathbb{R}^d$. The goal is the find vectors $Y_1, \dots, Y_n \in \mathbb{R}^s$ that serve as a good representation of the original vectors. The process of t-SNE starts with defining the following probability masses:

$$p_{j|i} = \frac{\exp(-\|X_i - X_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|X_i - X_k\|^2 / 2\sigma_i^2)}, \quad (1.1)$$

where σ_i are given constants. For each pair of data points X_i and X_j , we assign a weight to that pair, namely the Gaussian function $\exp(-\|X_i - X_j\|^2 / 2\sigma_i^2)$. We call such function the *input kernel*. In this paper, we are looking at generalized kernels. Let $\mathbf{Ker_I}(x, y, \sigma) : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^+$ be such that $\mathbf{Ker_I}(x, y, \sigma) = \mathbf{Ker_I}(y, x, \sigma)$. We replace the probability mass in (1.1) with the following

$$p_{j|i} = \frac{\mathbf{Ker_I}(X_i, X_j, \sigma_i)}{\sum_{k \neq i} \mathbf{Ker_I}(X_i, X_k, \sigma_i)}. \quad (1.2)$$

σ_i satisfies that for each $i = 1, \dots, n$, that

$$-\sum_{j=1, j \neq i}^n p_{j|i} \log p_{j|i} = \log \mathbf{Prep} =: \log(n\rho). \quad (1.3)$$

$\mathbf{Prep}$ is called the perplexity of the model and $\rho \in (0, 1)$ is a given constant. The purpose of σ_i is to make sure that for each i , the probability distribution induced by $p_{j|i}$ has the same entropy across all i . Clearly, $p_{j|i}$ is not symmetric. We then define $p_{ii} = 0$ and for $1 \leq i \neq j \leq n$,

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n} = \frac{1}{2n} \left(\frac{\mathbf{Ker_I}(X_i, X_j, \sigma_i)}{\sum_{k \neq i} \mathbf{Ker_I}(X_i, X_k, \sigma_i)} + \frac{\mathbf{Ker_I}(X_i, X_j, \sigma_j)}{\sum_{k \neq j} \mathbf{Ker_I}(X_j, X_k, \sigma_j)} \right). \quad (1.4)$$

Obviously,

$$\sum_{\substack{j=1 \\ j \neq i}}^n p_{j|i} = 1, \quad \sum_{1 \leq i \neq j \leq n} p_{ij} = 1. \quad (1.5)$$

As for the output vectors $Y_1, \dots, Y_n \in \mathbb{R}^s$, set for $1 \leq i \neq j \leq n$,

$$q_{ij} = \frac{\mathbf{Ker}_O(Y_i, Y_j)}{\sum_{k \neq l} \mathbf{Ker}_O(Y_k, Y_l)}. \quad (1.6)$$

The weight we assign to each pair of output data points Y_i and Y_j , a.k.a. a function $\mathbf{Ker}_O(Y_i, Y_j) : \mathbb{R}^s \times \mathbb{R}^s \rightarrow \mathbb{R}^+$, is called the output kernel. In the original algorithm, $\mathbf{Ker}_O(Y_i, Y_j) = (1 + \|Y_i - Y_j\|^2)^{-1}$. Denote $X = (X_1, \dots, X_n)$ and $Y = (Y_1, \dots, Y_n)$ the input and output data set. The goal of t-SNE is to find Y that minimizes the relative entropy

$$L_{n,\rho}(X, Y) = \sum_{1 \leq i \neq j \leq n} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right). \quad (1.7)$$

More precisely, the output of t-SNE Y^* is defined as

$$Y^* = \arg \min_{Y \in (\mathbb{R}^s)^n} L_n(X, Y). \quad (1.8)$$

In the expression of $L_n(X, Y)$, all p_{ij} are given and q_{ij} is a function of Y_i and Y_j . Therefore, Y^* can be found simply by using gradient descent. This is no different from the original version of t-SNE. However, it is the mathematical properties and implications of t-SNE that we are interested in. Towards this end, let $\mathcal{M}(\mathbb{R}^d)$ be the space of Borel probability measures on $\mathbb{R}^d$ and $\mu_X \in \mathcal{M}(\mathbb{R}^d)$ be a probability measure. Suppose all input $X = (X_1, \dots, X_n)$ are i.i.d., independent random variables with probability measure μ_X . Meanwhile, let $Y^* = (Y_1, \dots, Y_n)$ be the output of the t-SNE algorithm. Consider the following empirical measure:

$$\mu_n := \frac{1}{n} \sum_{i=1}^n \delta(X_i, Y_i) \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s). \quad (1.9)$$

We are interested in its limiting measure as $n \rightarrow \infty$. To finish the setup, there are a few more remaining quantity we would like to introduce. Let $\mu \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s)$. For a given kernel $\mathbf{Ker}_I(x, y)$, define $F_{\rho,\mu}(x, \sigma) : \mathbb{R}^d \times \mathbb{R}^+ \rightarrow \mathbb{R}$ as

$$F_{\rho,\mu}(x, \sigma) = \int \frac{\mathbf{Ker}_I(x, x', \sigma)}{\int \mathbf{Ker}_I(x, x', \sigma) d\mu(x')} \log \left(\frac{\mathbf{Ker}_I(x, x', \sigma)}{\int \mathbf{Ker}_I(x, x', \sigma) d\mu(x')} \right) d\mu(x') + \log \rho. \quad (1.10)$$

Note that the last s variables in the integrands stay constant. It is not hard to see that $F_{\rho,\mu}(x, \sigma)$ is the continuous version of the entropy. Next, we extend the definition of σ_i by defining $\sigma_{\rho,\mu}^* : \mathbb{R}^d \rightarrow \mathbb{R}$ the unique solution of the equation

$$F_{\rho,\mu}(x, \sigma) = 0. \quad (1.11)$$

The introduction of σ^* aligns with the procedure of finding σ_i that controls the entropy to be the same for all i . Of course, the existence and uniqueness of σ^* is not trivial and will be discussed in section 4. For the symmetric probability mass, let $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$ be a given function and $p_\psi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be defined by

$$p_\psi(x, x') = \frac{1}{2} \left(\frac{\mathbf{Ker}_I(x, x', \psi(x))}{\int \mathbf{Ker}_I(x, x', \psi(x)) d\mu(x')} + \frac{\mathbf{Ker}_I(x, x', \psi(x'))}{\int \mathbf{Ker}_I(x, x', \psi(x')) d\mu(x)} \right). \quad (1.12)$$

For the extension on the output side, let $q : \mathbb{R}^s \times \mathbb{R}^s \rightarrow \mathbb{R}$ be given by

$$q(y, y') = \frac{\mathbf{Ker}_O(y, y')}{\iint \mathbf{Ker}_O(y, y') d\mu(y) d\mu(y')}. \quad (1.13)$$

Based on the construction of p and q , define

$$I_\rho(\mu) = \iint p_{\sigma_{\rho,\mu}^*}(x, x') \log \left(\frac{p_{\sigma_{\rho,\mu}^*}(x, x')}{q(y, y')} \right) d\mu(x, y) d\mu(x', y'). \quad (1.14)$$

Recall $\mu_X \in \mathcal{M}(\mathbb{R}^d)$ is an arbitrary given probability measure. We set $\mathcal{P}_X$ to be the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^s$ whose marginal coincides with μ_X . Formally speaking,

$$\mathcal{P}_X := \{\mu \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s) : \text{the } \mathbb{R}^d \text{ marginal of } \mu \text{ is } \mu_X\}. \quad (1.15)$$

Lastly, define

$$\tilde{\mathcal{P}}_X := \left\{ \mu \in \mathcal{P}_X : \int (p_{\sigma_{\rho,\mu}^*}(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} \mathbf{Ker}_O(y, y')}{\mathbf{Ker}_O(y, y')} d\mu(x', y') = 0, \mu(x, y)\text{-a.e.} \right\}. \quad (1.16)$$

2 Main Results

The core results of this problem are the conditions we assume the input and output kernels to have such that the output of the generalized t -SNE algorithm converges to a limiting equilibrium measure. Towards this end, below are the constraints we impose on the input and output kernels.

Condition 1 (Input Kernel Conditions). *Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be some fixed function. The input kernel has the form of $\mathbf{Ker}_I(x_i, x_j, \sigma_i) = \exp(-w(\sigma_i \|x_j - x_i\|^\theta))$. Here, σ_i is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. (We can replace σ_i with $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^+$ in the continuous version of the kernel.) Define*

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \left| \lim_{n \rightarrow \infty} n^{-1/2} (\log n)^\theta \exp \left(4w \left(\left(\tilde{C} \sqrt{\log n} \right)^\theta \right) \right) = 0 \right. \right\}. \quad (2.1)$$

Then w satisfies that w satisfies:

1. $\forall t \geq 0, w'(t) > 0$ and $\exists M > 0$ s.t. $w'(t) \leq M$. $\lim_{t \rightarrow \infty} w(t) = \infty$;
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0$;
3. The following integral converges:

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty; \quad (2.2)$$

4. $\sup \mathcal{K}_w > 0$.

Condition 2 (Output Kernel Conditions). *The output kernel $\mathbf{Ker}_O(y, y') = k(\|y - y'\|)$ depends only on the distance between y and y' and that:*

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded;
2. g satisfies

$$\iint_{(0, \infty)^2} g(y, y') dy dy' < \infty; \quad (2.3)$$

3. k has bounded derivative;
4. $k'(0) = 0$.

We are ready to state the main results

Theorem 1. *Assume μ_X has a C^1 density $f(x)$ and compact support. Let $X_i \in \mathbb{R}^d$ be i.i.d random variables with probability measure μ_X . Then for any $\rho \in (0, 1)$,*

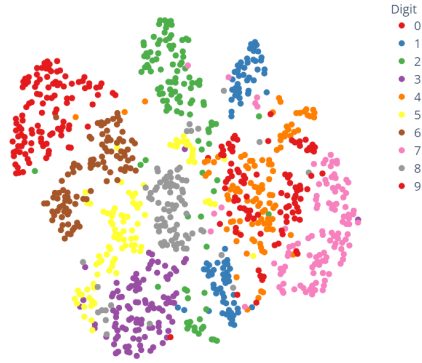
$$\lim_{n \rightarrow \infty} \inf_Y L_{n,\rho}(X, Y) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu). \quad (2.4)$$

Moreover, there exist a measure μ^ and a sub-sequence $\{n_k\}$ such that $\mu_{n_k} \rightarrow \mu^*$ weakly and $I_\rho(\mu^*) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu)$.*

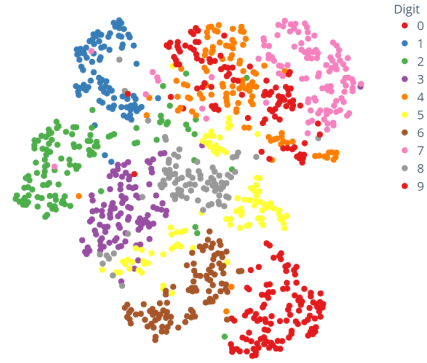
Theorem 2. *Under the same assumptions of theorem 1, μ^* has compact support.*

3 Intuitive Insights and Simulations

[Yi: TODO: Come back and finish this section when the rest of the paper is complete.]



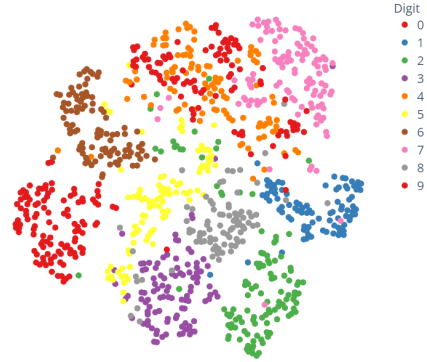
(a) Original t -SNE



(b) Polynomial Input Kernel



(c) Mixed Input Kernel



(d) Custom Input&Output Kernels

Figure 1: t -SNE running with different combinations of input and output kernels on the MNIST data set. All three custom kernel combinations are able to perform on par with the original algorithm.

4 Existence and Uniqueness of σ^*

5 Uniform Convergence of σ^*

6 Proofs of the Main Results

7 Extra Technical Proofs

References

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